LCCDE and convolution

Answers and worked solutions · Day 04 · Tue 27 Oct · pack D2

5. Practice

From your course files

T2 Q4 (tutorial).

> *Hint: simplify $h_2 - h_3 * h_4$ sample by sample before convolving.*
>
> **(a)** Cascade gives convolution and parallel paths add (the lower path enters the junction with a minus sign):
>
> $$h(n) = h_1(n) * \big(h_2(n) - h_3(n) * h_4(n)\big).$$
>
> **(b)** Using $x(n) * \delta(n-d) = x(n-d)$:
>
> $$h(n) = \{\underset{\uparrow}{2}, 1, 2\} * \big((n+1)u(n) - (n-1)u(n-2)\big) = \{\underset{\uparrow}{2}, 1, 2\} * \big(u(n) + u(n-1)\big)$$
>
> $$= 2\big(u(n)+u(n-1)\big) + \big(u(n-1)+u(n-2)\big) + 2\big(u(n-2)+u(n-3)\big) = \mathbf{2u(n) + 3u(n-1) + 3u(n-2) + 2u(n-3)}$$
>
> which is $\{2, 5, 8, 10, 10, \ldots\}$. (i) **Causal**, since $h(n) = 0$ for $n < 0$. (ii) **Unstable**, since $\sum_k |h(k)| = \infty$ ($h(n) \to 10$). See also Example 4.

T3 Q2 (tutorial).

> *Hint: substitute x directly. There is no feedback.*
>
> (a) There are no past outputs in the equation, so there is no feedback: **the characteristic modes are zero** (there are none).
>
> (b) By direct substitution:
>
> $$y[n] = \tfrac{1}{4}\big(5^{-n}u[n] + 2\cdot 5^{-(n-1)}u[n-1] + 5^{-(n-2)}u[n-2]\big) = \mathbf{\tfrac{1}{4}5^{-n}\big(u[n] + 10u[n-1] + 25u[n-2]\big)}$$

T3 Q4 (tutorial).

> (a) $\lambda^2 + \lambda + \frac{1}{4} = (\lambda + \frac{1}{2})^2 = 0$, a repeated root $\lambda = -\frac{1}{2}$. The modes are $(-2)^{-n}$ and $n(-2)^{-n}$, so $y_h[n] = C_1(-2)^{-n} + C_2 n(-2)^{-n}$.
>
> (b) Try $y_p[n] = K2^{-n}$: $K(1+2+1) = 1$, so $K = 0.25$. The ICs enter **only** through the recursion:
>
> $$y[0] = x[0] - y[-1] - \tfrac{1}{4}y[-2] = 1 - (-1) - \tfrac{1}{4}(2) = 1.5, \qquad y[1] = x[1] - y[0] - \tfrac{1}{4}y[-1] = 0.5 - 1.5 - \tfrac{1}{4}(-1) = -0.75.$$

Fit $y[n] = C_1(-2)^{-n} + C_2 n(-2)^{-n} + 0.25 \cdot 2^{-n}$: $n = 0$ gives $C_1 = 1.25$; $n = 1$ gives $-0.625 - 0.5C_2 + 0.125 = -0.75$, so $C_2 = 0.5$.

$$\mathbf{y[n] = 1.25(-2)^{-n} + 0.5\,n(-2)^{-n} + 0.25 \cdot 2^{-n}, \quad n \geq 0}$$

Note: the printed solution's line for $y[1]$ reads "$0.5 - 2.5 - \frac{1}{4}(-1)$". It should be $0.5 - 1.5 - \frac{1}{4}(-1)$, because $y[0] = 1.5$. The printed result, -0.75, is correct, and so is the final answer (verified in Python to $n = 7$).

T3 Q5 (tutorial).

(a) No feedback, so substitute $x[n] = \delta[n]$:
$h[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 4\delta[n-4] = \{\underset{\uparrow}{\mathbf{1}}, \mathbf{2}, \mathbf{3}, \mathbf{0}, \mathbf{4}\}$. Don't skip the zero at $n = 3$.

(b) $\lambda^2 + 0.2\lambda - 0.08 = 0$ gives $\lambda = 0.2, -0.4$, so $h[n] = C_1(0.2)^n + C_2(-0.4)^n$ for $n \geq 0$. Recursion: $h[0] = 1 = C_1 + C_2$ and $h[1] = -0.2h[0] = -0.2 = 0.2C_1 - 0.4C_2$. So $C_1 = \frac{1}{3}$, $C_2 = \frac{2}{3}$ and $\mathbf{h[n] = \left(\frac{1}{3}(0.2)^n + \frac{2}{3}(-0.4)^n\right)u[n]}$.

(c) $\lambda^2 - 2\lambda + 1 = 0$ gives $\lambda = 1$ repeated, so the modes are 1 and n: $h[n] = C_1 + C_2 n$. $h[0] = C_1 = 1$ and $h[1] = C_1 + C_2 = 2$, so $\mathbf{h[n] = (1 + n)u[n]}$.

T3 Q6 (tutorial).

(a) Long multiplication (Playbook E). x starts at $n = 0$ and h at $n = -2$, so y starts at $n = -2$:

$$y[n] = \{2 \;\; -3 \;\; \underline{-3} \;\; -5 \;\; 6 \;\; 5 \;\; 10 \;\; 12\}$$

(b)

$y[n] = \sum_k h[k]x[n-k] = \sum_{k=0}^{n} 2^{-k}3^{n-k} = 3^n \sum_{k=0}^{n} 6^{-k} = \mathbf{3^n\left(\frac{1 - 6^{-(n+1)}}{1 - 6^{-1}}\right)} = ($

.

(c) The same with $e^{j\omega_0}$ in place of 3: $\mathbf{y[n] = e^{j\omega_0 n}\left(\frac{1 - (2e^{j\omega_0})^{-(n+1)}}{1 - (2e^{j\omega_0})^{-1}}\right)u[n]}$. Further simplification is not necessary.

(d) $\cos(\omega_0 n) = \mathcal{R}\{e^{j\omega_0 n}\}$ and h is real, so the output is the real part of (c):
$y[n] = \mathcal{R}\left\{e^{j\omega_0 n}\left(\frac{1 - (2e^{j\omega_0})^{-(n+1)}}{1 - (2e^{j\omega_0})^{-1}}\right)\right\}u[n]$. (Alternatively write $\cos(\omega_0 n) = \frac{1}{2}(e^{j\omega_0 n} + e^{-j\omega_0 n})$ and use (c) twice.)

New questions

Q1 (easy) [2].

The output starts at $n = -1 + 0 = -1$ and has $3 + 4 - 1 = 6$ samples.

	$n = -1$	**0**	**1**	**2**	**3**	**4**
$1 \cdot h$	3	0	1	2		
$2 \cdot h$		6	0	2	4	
$-1 \cdot h$			-3	0	-1	-2
y	3	6	-2	4	3	-2

$$\mathbf{y(n)} = \{\mathbf{3}, \underset{\uparrow}{\mathbf{6}}, \mathbf{-2}, \mathbf{4}, \mathbf{3}, \mathbf{-2}\}$$

Check: $\sum y = 12 = (2)(6)$.

Q2 [3].

$y(n) = \sum_k x(k)h(n-k) = \sum_k 0.5^{n-k}$ over $0 \le k \le 3$ (from x) and $k \le n$ (from $u(n-k)$).

- $n < 0$: there is no overlap, so $y = 0$.
- $0 \le n \le 3$: $y(n) = 0.5^n \sum_{k=0}^{n} 2^k = 0.5^n(2^{n+1} - 1) = 2 - 0.5^n$.
- $n \ge 4$: $y(n) = 0.5^n \sum_{k=0}^{3} 2^k = 15(0.5)^n$.

$$\mathbf{y(n)} = \begin{cases} 0 & n < 0 \\ 2 - 0.5^n & 0 \le n \le 3 \\ 15(0.5)^n & n \ge 4 \end{cases}$$

Values: $1, 1.5, 1.75, 1.875, 0.9375, \ldots$ (Python-checked). At the joint, $n = 4$ gives $15/16 = 0.9375$, which matches $y(3) \cdot 0.5 = 0.9375$ (the pulse has ended, so the output just decays).

Q3 [3].

The mode is $\lambda = 0.5$. With $x = \delta$ and zero ICs: $h(0) = 1$ and $h(1) = 0.5(1) - 1 = -0.5$. For $n \ge 2$, $h(n) = 0.5h(n-1)$.

Here $m = k = 1$, so the mode form holds only from $n = 1$: $h(n) = C(0.5)^n$ for $n \ge 1$. From $h(1)$, $C = -1$. Write $h(n) = D\delta(n) - 0.5^n u(n)$. Then $n = 0$ gives $D - 1 = 1$, so $D = 2$.

$$\mathbf{h(n) = 2\delta(n) - (0.5)^n u(n) = \{\underset{\uparrow}{1}, -0.5, -0.25, -0.125, \ldots\}}$$

It is causal. It is IIR. It is stable: $\sum |h| = 1 + \sum_{n\geq 1} 0.5^n = 2 < \infty$ (equivalently $|\lambda| = 0.5 < 1$).

Alternative: by linearity and time invariance, $h = g(n) - g(n-1)$ with $g = 0.5^n u(n)$. This gives the same sequence.

Q4 [4].

1. $\lambda^2 - \lambda + 0.25 = (\lambda - 0.5)^2 = 0$, a repeated root, so $y_h = C_1 0.5^n + C_2 n 0.5^n$.
2. Try $y_p = K3^{-n}$. Since $1/3$ is not a root this is fine: $K(1 - 3 + 0.25 \cdot 9) = 1$, so $0.25K = 1$ and $K = 4$.
3. Recursion: $y(0) = 1$. $y(1) = \frac{1}{3} + 1 = \frac{4}{3}$.
4. $n = 0$: $C_1 + 4 = 1$, so $C_1 = -3$. $n = 1$: $0.5(-3) + 0.5C_2 + \frac{4}{3} = \frac{4}{3}$, so $C_2 = 3$.

$$\mathbf{y(n) = (-3(0.5)^n + 3n(0.5)^n + 4(1/3)^n)u(n)}$$

Check $n = 2$: the recursion gives $\frac{1}{9} + \frac{4}{3} - \frac{1}{4} = 1.19444$, and the formula gives $-0.75 + 1.5 + 0.4444 = 1.19444$.

Q5 [3].

a. $\lambda^2 - \lambda + 0.5 = 0$, so $\lambda = \frac{1 \pm j}{2} = \frac{1}{\sqrt{2}} e^{\pm j\pi/4}$. The modes are $(1/\sqrt{2})^n \cos(\pi n/4)$ and $(1/\sqrt{2})^n \sin(\pi n/4)$.

b. Recursion: $h(0) = 1$ and $h(1) = h(0) = 1$. Let $h(n) = (1/\sqrt{2})^n [A\cos(\pi n/4) + B\sin(\pi n/4)]$. At $n = 0$, $A = 1$. At $n = 1$, $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1 + B) = 1$, so $B = 1$.

$$\mathbf{h(n) = \left(\frac{1}{\sqrt{2}}\right)^n \left[\cos\frac{\pi n}{4} + \sin\frac{\pi n}{4}\right] u(n) = \{1, 1, 0.5, 0, -0.25, -0.25, \ldots\}}$$

Check $h(2)$: the recursion gives $1 - 0.5 = 0.5$, and the formula gives $0.5[0 + 1] = 0.5$.

c. **Stable**: $|\lambda| = 0.707 < 1$, so h decays geometrically. Numerically $\sum |h| \approx 3.33$.

Q6 (chained) [7].

a. There is one mode, $\lambda = 0.5$, and $h_1(0) = 1$, so $h_1(n) = (0.5)^n u(n)$.

b. $h = h_1 * h_2 = \dfrac{0.5^{n+1} - 0.25^{n+1}}{0.5 - 0.25} u(n)$, so

$$\mathbf{h(n) = (2(0.5)^n - (0.25)^n)u(n).}$$

Check: $h(0) = 1 = h_1(0)h_2(0)$.

c. The recursion with $x = \delta$ gives $h(0) = 1$ and $h(1) = 0.75$. The formula gives $2 - 1 = 1$ and $1 - 0.25 = 0.75$. The characteristic equation $\lambda^2 - 0.75\lambda + 0.125 = (\lambda - 0.5)(\lambda - 0.25)$ has one root from each stage. A cascade's modes are the union of the stages' modes.

d. $y = h * x = 2[0.5^n u * 0.5^n u] - [0.25^n u * 0.5^n u]$. The first term has equal bases, so it is $2(n+1)0.5^n$. The second is $\dfrac{0.5^{n+1} - 0.25^{n+1}}{0.25} = 2(0.5)^n - 0.25^n$. Subtracting:

$$\mathbf{y(n) = (2n(0.5)^n + (0.25)^n)u(n)}$$

Check: $y(0) = 1 = h(0)x(0)$.
$y(1) = 1 + 0.25 = 1.25 = h(0)x(1) + h(1)x(0) = 0.5 + 0.75$. All values were Python-checked.

6. Quick recall check

Question

1. Only the y coefficients: $\lambda^k + a_1\lambda^{k-1} + \cdots + a_k = 0$. An FIR LCCDE has no feedback, so it has no modes ("the modes are zero").
2. $C_1(-\frac{1}{2})^n + C_2 n(-\frac{1}{2})^n$, written in the course as $C_1(-2)^{-n} + C_2 n(-2)^{-n}$.
3. $\dfrac{a^{n+1} - b^{n+1}}{a - b}u(n)$; and $(n+1)a^n u(n)$.
4. No. $\sum |h(k)| = \infty$ because a unit-magnitude oscillation never decays.
5. It starts at $n = -3$, with $4 + 5 - 1 = 8$ samples ($n = -3$ to 4).